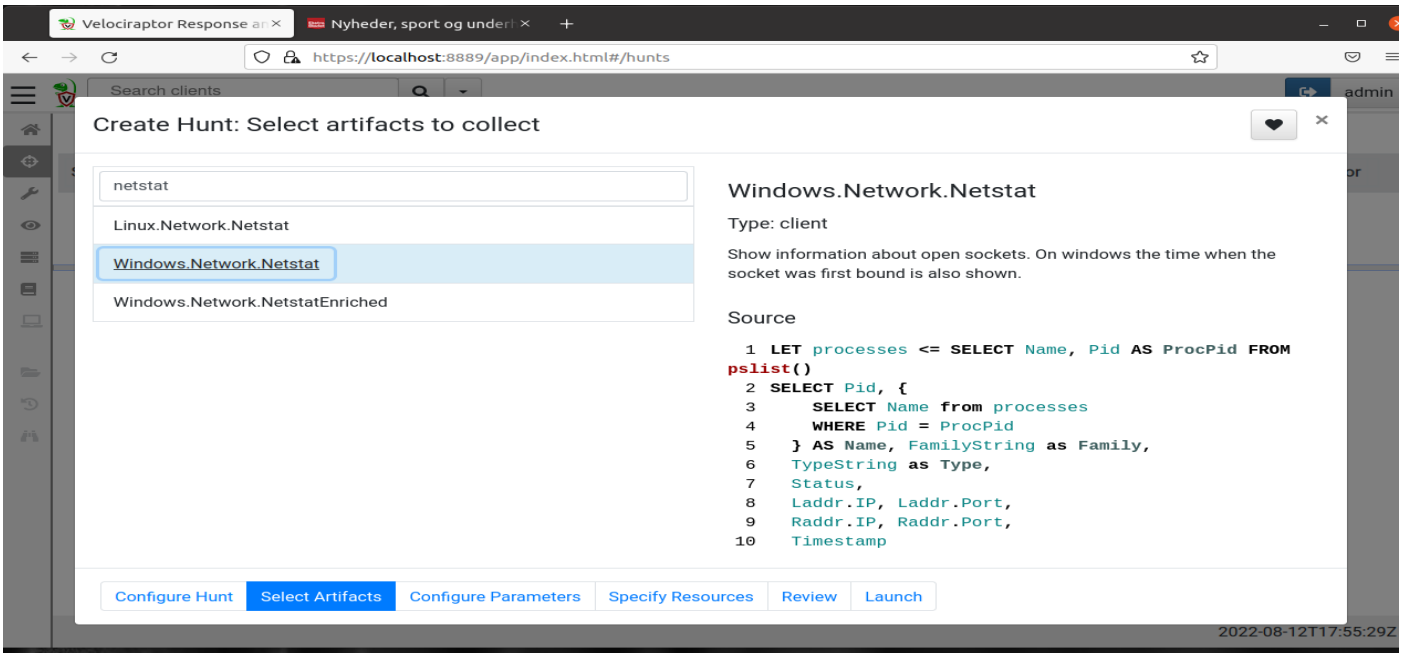
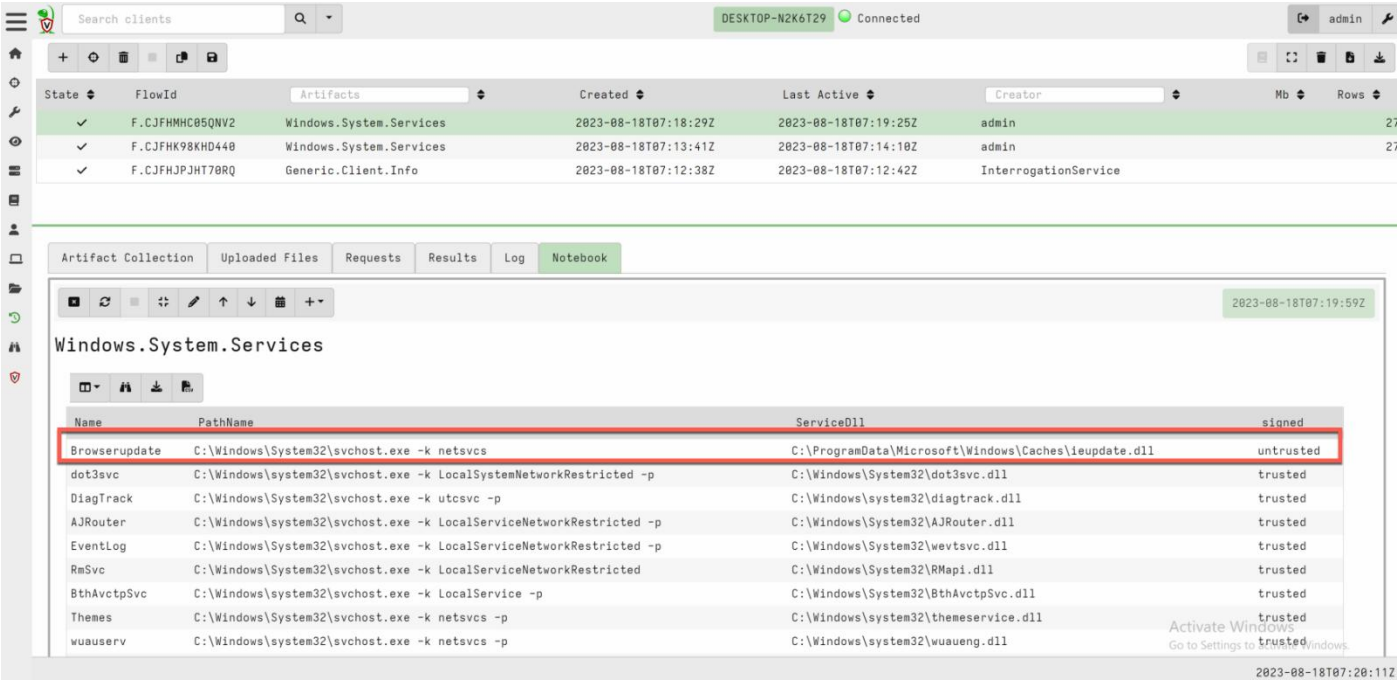
1. Volatile Data Collection (Velociraptor)

Volatile data was collected from a Windows virtual machine using Velociraptor. The query SELECT * FROM netstat was executed to capture active network connections. The results included details such as protocol, local address, foreign address, and connection state. The collected data was saved in CSV format for further analysis.



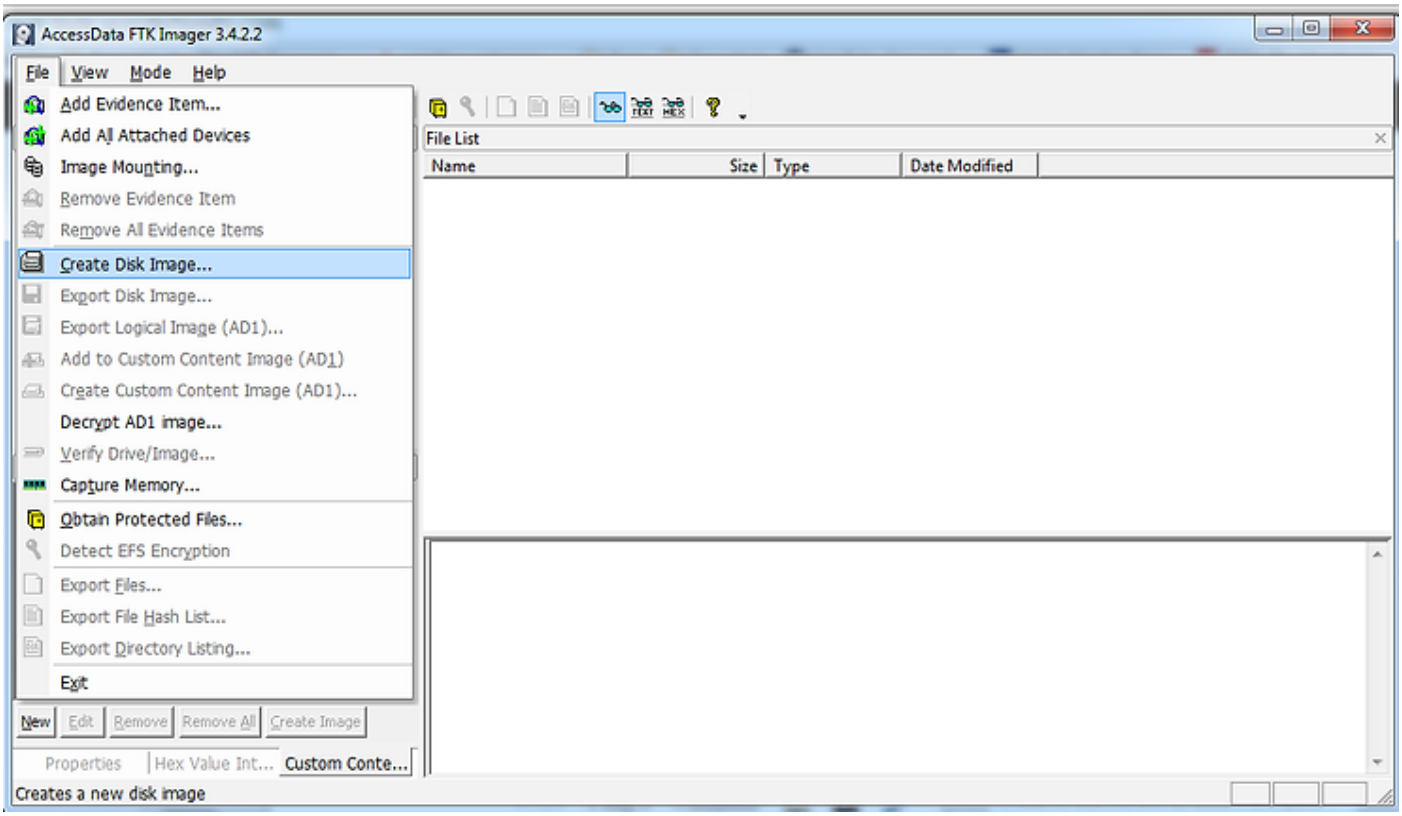
Source Netstat

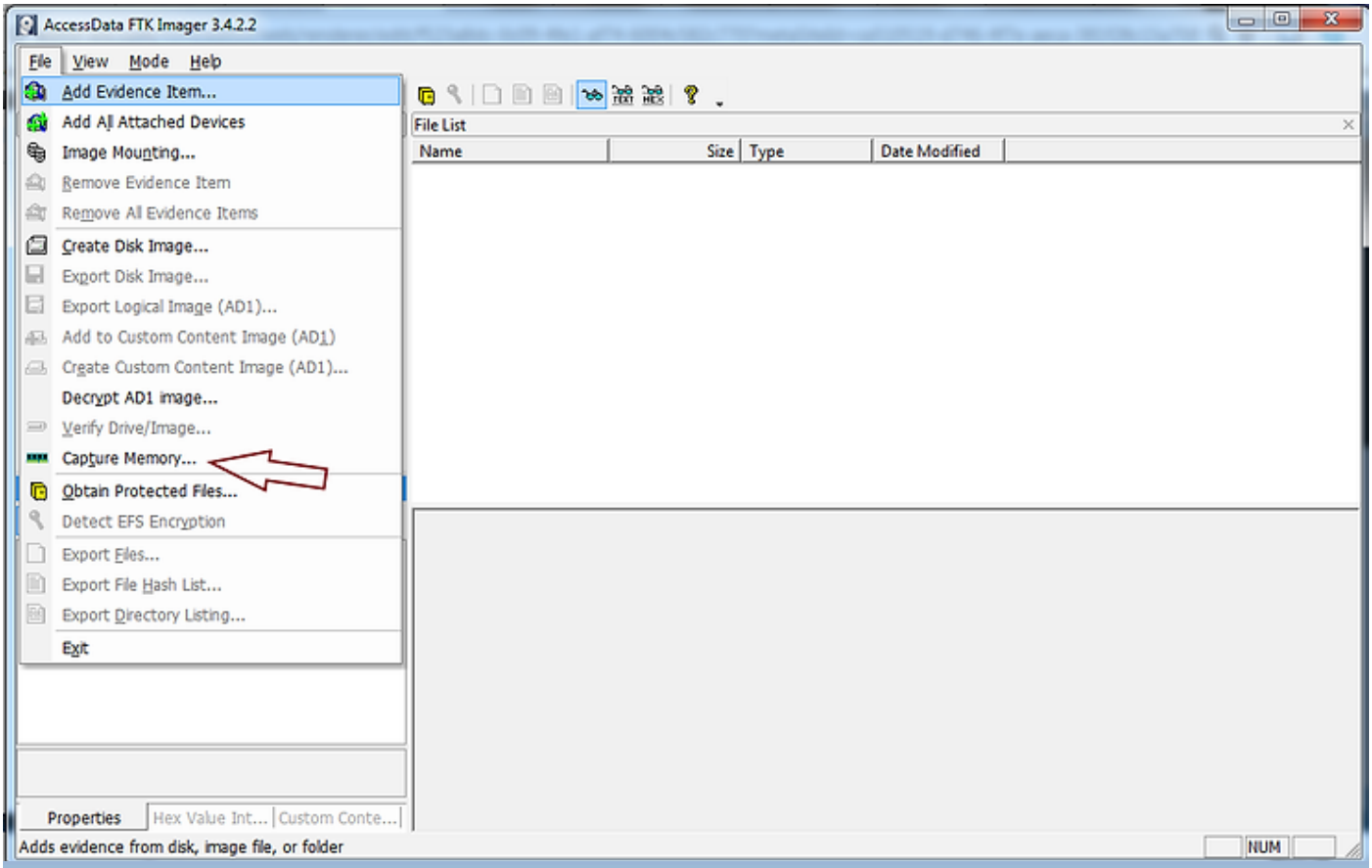
```
LET VerifiedRegex <= SELECT Regex
      FROM parse_csv(filename=AuthenticodeVerifiedMap, accessor="data")
      WHERE Choice=AuthenticodeVerified LIMIT 1
LET StatusRegex <= SELECT Regex
      FROM parse_csv(filename=StatusMap, accessor="data")
      WHERE Choice=Status LIMIT 1
LET FamilyRegex <= SELECT Regex
      FROM parse_csv(filename=FamilyMap, accessor="data")
      WHERE Choice=Family LIMIT 1
LET TypeRegex <= SELECT Regex
      FROM parse_csv(filename=TypeMap, accessor="data")
      WHERE Choice=Type LIMIT 1

LET process <= SELECT Pid as PsId,
      Ppid,
      Name,
      CommandLine,
      Exe,
      Hash,
      Authenticode,
      Username
  FROM Artifact.Windows.System.Pslist(
      DISABLE_DANGEROUS_API_CALLS=DISABLE_DANGEROUS_API_CALLS)
  WHERE Name =~ ProcessNameRegex
      AND Exe =~ ProcessPathRegex
      AND CommandLine =~ CommandLineRegex

LET results = SELECT Pid,
          { SELECT Ppid FROM process WHERE PsId = Pid } as Ppid,
          { SELECT Name FROM process WHERE PsId = Pid } as Name,
```

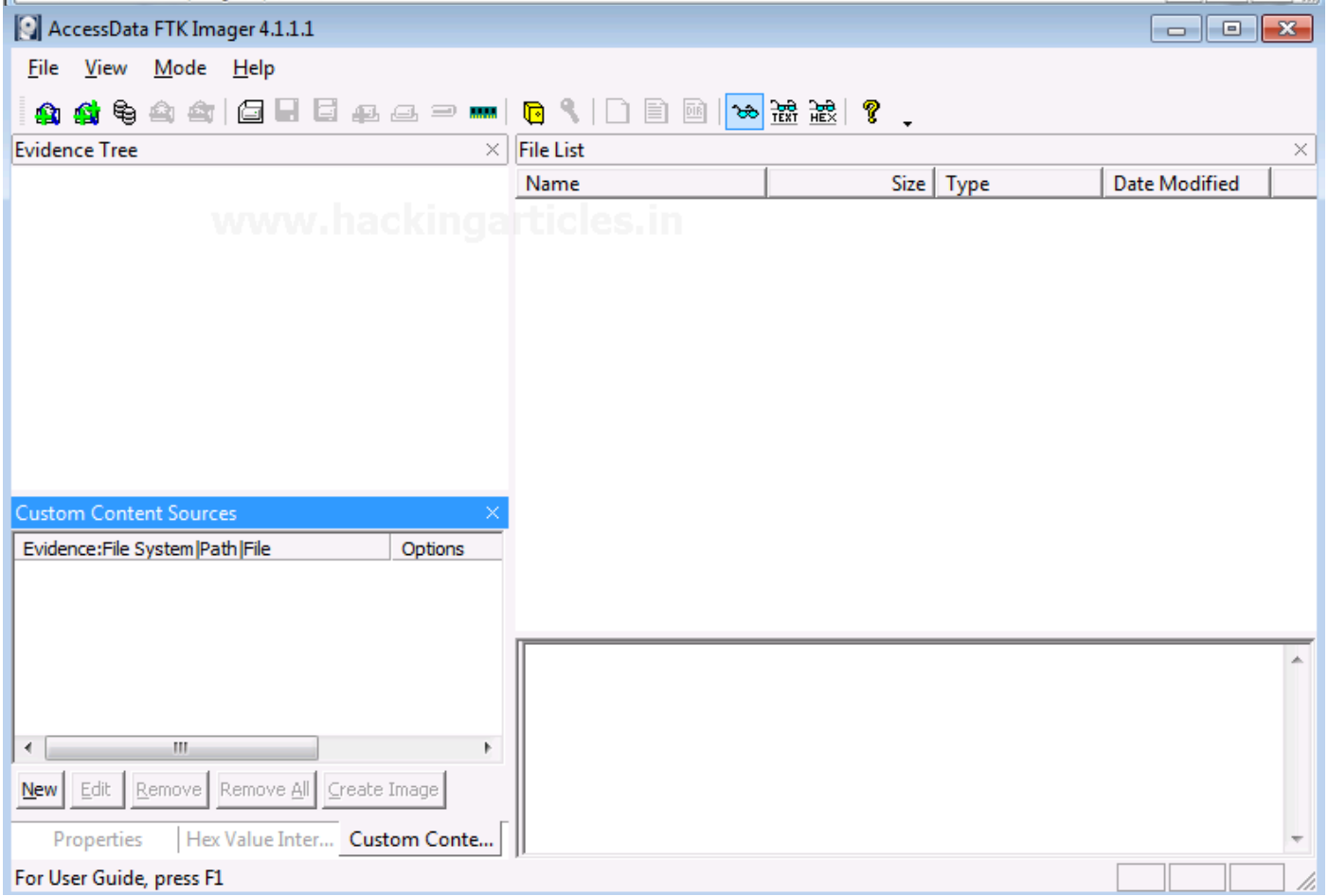


2. Evidence Collection and Hashing

A memory dump was collected using the Velociraptor artifact Artifact.Windows.Memory.Acquisition. This process captured the system's RAM for forensic analysis. To ensure data integrity, a SHA256 hash value was generated for the memory dump file using a hashing tool. This helps verify that the evidence has not been altered.







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C:\WINDOWS\system32\cmd.exe

```
Microsoft Windows [Version 10.0.19042.985]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ac>d:

D:\>cd \FFmpeg\ffmpeg-4-4-2021-05-26

D:\FFmpeg\ffmpeg-4-4-2021-05-26>dir/b
ffmpeg-4.4-full_build.7z

D:\FFmpeg\ffmpeg-4-4-2021-05-26>certutil -hashfile ffmpeg-4.4-full_build.7z SHA256
SHA256 hash of ffmpeg-4.4-full_build.7z:
3b18ea042dcc0865435f1a03d2c363f7bfd7cf8b52a3081253bc9d6afd1d823f
CertUtil: -hashfile command completed successfully.

D:\FFmpeg\ffmpeg-4-4-2021-05-26>
```

Command Prompt

```
Microsoft Windows [Version 10.0.18362.778]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\SeanKOShea>cd c:\baseball2017

c:\baseball2017>for %F in (*) do @certutil -hashfile "%F" MD5 | find /v "hashfile command completed successfully" >>list.txt

c:\baseball2017>
```

pwsh in denelon

```
denelon winget hash --help
Windows Package Manager v1.5.1572
Copyright (c) Microsoft Corporation. All rights reserved.

Computes the hash of a local file, appropriate for entry into a manifest.  It can also compute the hash of the signature fil
e of an MSIX package to enable streaming installations.

usage: winget hash [-f] <file> [<options>]

The following arguments are available:
  -f,--file                 File to be hashed

The following options are available:
  -m,--msix                 Input file will be treated as msix; signature hash will be provided if signed
  -?,--help                 Shows help about the selected command
  --wait                    Prompts the user to press any key before exiting
  --logs,--open-logs        Open the default logs location
  --verbose,--verbose-logs  Enables verbose logging for winget
  --disable-interactivity   Disable interactive prompts

More help can be found at: https://aka.ms/winget-command-hash
denelon                                                                                    in pwsh at 09:33:41
```

3. Chain of Custody Documentation

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Dump	SOC Analyst	2025-08-18	3f2a9c8e7b1d4e6f9a0c123456789abcdef123456789abcdef123456789abcdef

4. Analysis

The network connections collected showed active sessions on the system, which can help identify suspicious communication. The memory dump provides detailed information about running processes and potential malware. Hashing ensures the integrity of the evidence, which is important for forensic investigation.

5. Conclusion

The evidence was successfully collected and preserved following proper procedures. The use of hashing ensured data integrity, and maintaining a chain of custody helps in tracking the handling of evidence. Further analysis can be performed to detect any malicious activity.